Akoizo Original Practice - Form 2 - Answer key

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) If airspeed doubles with all else equal, how does lift change?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. It halves

B. It stays the same

C. It becomes four times larger

D. It doubles

Correct: C

2. (2 points) How many test flights does a complete log require?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. 10 or more

B. 5

C. 20

D. 1 per motor

Correct: A

3. (2 points) A glide shows repeated nose-up, nose-down stalls. What is likely?

A. Too much decalage or CG too far back

B. CG too far forward

C. Not enough dihedral

D. Propeller too small

Correct: A

4. (2 points) A plane hits the ceiling early in the flight. What is the best first change?

A. Back off turns to reduce launch torque

B. Wind more turns

C. Use a thicker motor

D. Add camber

Correct: A

5. (4 points) Why is a complete flight log worth building?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

answer: It multiplies every flight by 1.2 instead of 1.1 or 1.0, and its data predicts the best winding for competition. (4 points).

6. (4 points) Why add ballast where it improves balance rather than anywhere?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

answer: Ballast changes the CG; placing it to move the CG toward the target fixes balance while meeting the minimum mass. (4 points).

7. (4 points) How does propeller choice affect flight time directly?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

answer: It sets how fast the motor unwinds; a well-matched slow propeller spends turns more slowly while still giving enough thrust. (4 points).

8. (4 points) How does camber help an indoor wing?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

answer: It raises the lift coefficient at low speed, so the wing supports the plane while flying slowly. (4 points).

9. (4 points) Name three useful extra log parameters.

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

answer: Torque at landing, estimated peak altitude, and propeller pitch; others include motor mass and ceiling height. (4 points).

10. (4 points) Explain how dihedral rolls a plane level.

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

answer: When one wing drops, the plane slips toward it and the lower wing meets the air at a higher angle, making more lift and rolling it back. (4 points).

11. (6 points) Solve this case and explain your reasoning: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?

reasoning: Safe turns = 1,400 x 0.92 about 1,290 turns.
Cranks = 1,290 ÷ 15 about 86 cranks.
Check torque at that count and record both in the log. (3 points).

conclusion: Wind about 86 cranks, about 1,290 turns, and confirm the torque. (3 points).

12. (2 points) Which team scores highest?

Supplied synthetic observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. Team B

B. Team A

C. Team C

Correct: A

13. (2 points) What caused the drop in performance?

Supplied synthetic observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. The motor lost mass

B. The ceiling was lower

C. Heat warped the wing, changing trim

Correct: C

14. (2 points) Define “Thrust” in the context of Wright Stuff.

answer: Forward force produced by the propeller. (2 points).

15. (2 points) Define “Lift coefficient (C_L)” in the context of Wright Stuff.

answer: Dimensionless measure of a wing’s lifting ability at a given angle. (2 points).

16. (2 points) Define “Peak altitude” in the context of Wright Stuff.

answer: Highest point of a flight; a useful log parameter. (2 points).

17. (2 points) Define “Decalage” in the context of Wright Stuff.

answer: Difference in wing and stabilizer incidence angles. (2 points).

18. (2 points) Define “Stretch-winding” in the context of Wright Stuff.

answer: Winding while the motor is stretched to fit more turns. (2 points).